White paper: DGP architecture choice and carryover power loss

2026-07-29 09:02 PDT

Summary of 'Three data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and why the choice changes power under carryover: a tutorial' (pmsimstats team, 2026-06-17).

Goal of the paper

Aggregated N-of-1 trials pool many single-patient on/off series to validate predictive biomarkers. Power for the biomarker-by-treatment interaction has no closed form in these designs, so it is obtained by Monte Carlo simulation. Every such simulation must encode how the biomarker moderates the treatment effect in its data-generating process (DGP), and there is more than one way to do so. That encoding is normally treated as an implementation detail.

The paper's goal is to show that it is not a detail but a substantive scientific assumption. Specifically, it aims to:

1. Formalize three DGP architectures as co-equal modeling choices.
2. Quantify, by simulation with the analysis model held fixed, how much each architecture's estimated power erodes as carryover half-life increases, across three within-subject designs.
3. Supply guidance on which architecture matches which biomarker biology, and on what simulation studies must report.

The work is motivated by a concrete discrepancy: the original Hendrickson et al. (2020) code used one encoding, a re-implementation in the same software project quietly used another, and the two gave different accounts of how much carryover costs.

The three architectures

- **Architecture A (direct mean moderation).** The biomarker scales the treatment effect in the conditional mean, $Y_{it} = \mu_0 + \beta_1 D_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$. The interaction is deterministic and lives in the first moment. This is the near-universal convention in the trial-simulation literature.
- **Architecture B (differential correlation, MVN).** Biomarker and biological response are drawn jointly from a multivariate normal whose BM-BR correlation is treatment-state dependent: c_{bm} on drug, $c_{bm}e^{-\lambda t_{sd}}$ off drug. The interaction is probabilistic and lives in the second moment. This is the Hendrickson N-of-1 approach.
- **Architecture C (combined).** Both channels act at once through independent parameters $c_{bm,A}$ (mean) and $c_{bm,B}$ (covariance); A and B are its single-channel limits, enforced algebraically rather than by convention.

Simulation setup

Identical configurations were run under each architecture, differing only in the BM-BR linkage. Designs: crossover (CO), Hybrid, and open-label with blinded discontinuation (OL+BDC). Primary run $N = 35$ per randomization path, $c_{bm} = \beta_{bm} = 0.45$, DGP carryover half-life $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks with a matched analysis half-life, 1,000 replicates per cell (Monte Carlo SE near 0.016), 36,000 fits, 100% convergence. The analysis model was fixed throughout: `nlme::lme` with `corCAR1` residual correlation and the interaction term `bm:Dbc`, where `Dbc` equals 1 on drug and $e^{-\lambda t_{sd}}$ off drug.

Results

Architecture A preserves power. Moving from no carryover to a one-week half-life costs only 1.9% relative power in CO, 1.4% in Hybrid, and 2.8% in OL+BDC (0.751 to 0.730). The biomarker proportionality is maintained at every occasion, so the interaction signal contracts in amplitude without losing identifiability.

Architecture B loses substantially, and design-dependently. The same carryover costs 3.0% in CO (not distinguishable from zero), 5.1% in Hybrid, and 30.6% in OL+BDC, where power falls from 0.777 to 0.539. The OL+BDC loss is roughly eleven times the matched Architecture A loss.

The gap is far beyond Monte Carlo error. A difference-of- differences z-test on the within-architecture losses gives $z = 7.64$ ($p < 10^{-13}$) at OL+BDC and $z = 3.96$ ($p < 10^{-4}$) at Hybrid; the CO contrast is null ($z = 0.29$), because the crossover design's within-subject AR(1) structure already dominates the correlation channel that Architecture B uses to carry the signal.

Mechanism. Both architectures suffer mean blurring, since carryover compresses the `Dbc` contrast. Architecture B suffers a second, unique attack: the BM-BR correlation that generates its second-moment signal decays exponentially off drug. Under Architecture A the drug-mediated and drug-independent components of BR variance move together, so their ratio, and hence identifiability, survives.

Architecture C is intermediate. Across the 3×3 grid of $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$ at $t_{1/2} = 1.0$, power increases strictly with each parameter, and the boundary cells reproduce the pure architectures exactly. The paper reads the interior cells as super-additive: Hybrid $(0.22, 0.22)$ reaches 0.299 against an additive-on-the-probability-scale benchmark of 0.157. At full signal the Hybrid design retains 0.870, exceeding either pure architecture alone.

Auxiliary checks. Type I error across the 18 null cells sat in [0.010, 0.074] (mean 0.043) with no architecture-specific inflation. An $N = 70$ run on OL+BDC preserved the B-versus-A ordering (9.3 versus 1.7 percentage points lost), compressed by ceiling saturation.

What follows for practice

The architecture encodes *why* the biomarker predicts response, so choosing one is a biological claim. Use Architecture A when the biomarker mechanically sets the size of the drug effect (renal clearance, receptor density, metabolizer genotype). Use Architecture B when the biomarker is only a proxy for a hidden responder state (resting blood pressure in PTSD, CRP or IL-6, polygenic risk scores). The motivating prazosin/PTSD application is an Architecture B story, which makes its carryover sensitivity a clinically relevant warning rather than an artifact.

Because the literature uses Architecture A almost exclusively, published power figures may be systematically optimistic for biomarkers that genuinely operate through a correlation channel. The distinction bites hardest in treatment-switching designs (crossover, N-of-1, basket, adaptive enrichment) and is immaterial in parallel-group designs with no off-drug phase. Recommendations: state which architecture a simulation adopts and why; when the mechanism is uncertain, sweep Architecture C across a mixing weight and report the range; treat Architecture B as the conservative default; and reduce exposure to the question by designing adequate washout.

Does Architecture C earn its place?

Architecture C is worth keeping, but for narrower reasons than the paper claims, and its current execution understates the case. The defensible arguments are two. First, a verification argument: the boundary cells ($c_{bm,A} = 0.45, c_{bm,B} = 0$) and $(0, 0.45)$ recover the pure architectures algebraically rather than by convention, so any C-based simulation carries a built-in correctness gate on the DGP implementation. Second, a design argument: when the mechanism is unknown, C with a mixing weight α is the only construct spanning the A-to-B continuum in one parameterization, which is what a sensitivity analysis requires. The power argument, by contrast, is weak as presented. Power is a convex function of effect size below 0.5, so additivity on the probability scale is the wrong benchmark: a normal approximation in which the two channels merely add on the effect-size scale already predicts 0.254 for the Hybrid $(0.22, 0.22)$ cell against 0.157 for the probability-scale benchmark and 0.299 observed, leaving a residual of roughly three Monte Carlo standard errors rather than the reported gap of nine. Three further caveats constrain what C can deliver. The $c_{bm,A}$ and $c_{bm,B}$ parameters are not separately identifiable from a fitted `bm:Dbc` coefficient, so C is a DGP-side construct that can be simulated but never estimated from trial data, which makes the recommendation to obtain independent evidence on each channel difficult to act on. The C grid uses $N = 35$ as a total rather than per path, so its cells are not readable against the headline tables. And the carryover comparison within the grid is reported only for the crossover design, which is precisely the design in which Section 3 found no architecture-by-carryover interaction; the informative panel would be OL+BDC. Net assessment: retain C as a scoped sensitivity device with the effect-size-matched parameterization ($c_{bm,A} = (1 - \alpha)c$, $c_{bm,B} = \alpha c$) that the paper defines but declines to use, rather than presenting it as a co-equal third architecture.

Scope and limitations

The paper is a tutorial and simulation study, not an empirical validation against trial data. Section 6's power-recovery strategies (on-drug-only analysis, exposure weighting, within-subject contrasts, two-stage random slopes) are qualitative hypotheses with no simulation support here; a companion paper addresses them. Architecture B is acknowledged as a correlation-level approximation to what would more faithfully be a finite-mixture responder model. The effect-size-scale recomputation in the preceding section is a normal-approximation check performed for this summary, not a re-simulation.

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